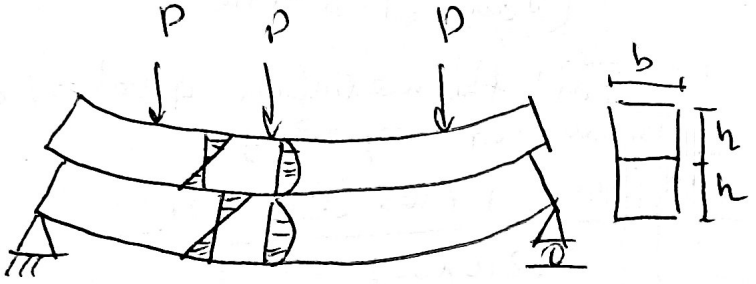
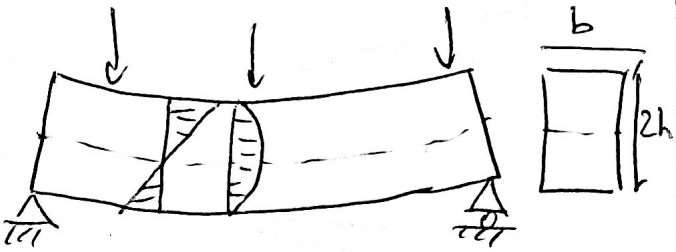
COMPOUND BEAMS

In order to increase the moment resistance under the effect of bending bars can be connected to eachother by wedges, bolts nuts and weld. The resulting beams are called compound beams



no connection between beams
free relative sliding
no shear occurs (it is assumed that friction is too low)

$$M_b = 2\left(\frac{\sigma_{em}\frac{1}{12}bh^3}{\frac{1}{2}h}\right) = \frac{1}{3}\sigma_{em}bh^2$$



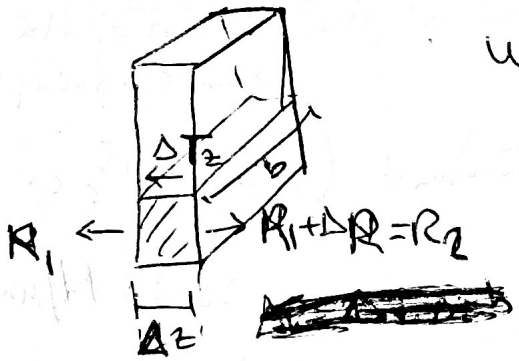
connected beams (work together)
Shear occurs at the interface

$$M_c = \frac{\sigma_{em}\frac{1}{12}b(2h)^3}{h} = \frac{2}{3}\sigma_{em}bh^2$$

$$\boxed{M_c = 2M_b}$$

Compound beam can carry two fold load.

Shear Flow



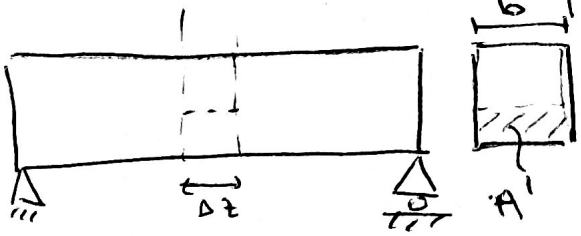
We know that $R_1 + \Delta T_z = R_2$ and $\Delta T_z = \Delta R$

$$\Delta T_z = \tau_{zy}\, b(y)\, \Delta z = \int_{A'} \Delta\tau_z \, dA'$$

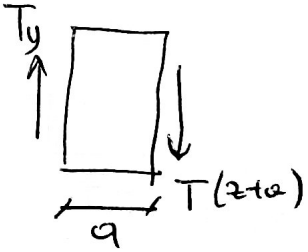
$$q = \frac{\Delta T_z}{\Delta z} = \tau_{zy}\, b(y) = \frac{T_y S_x}{I_x} = \int_A \frac{d\tau_z}{dz}\, dA$$

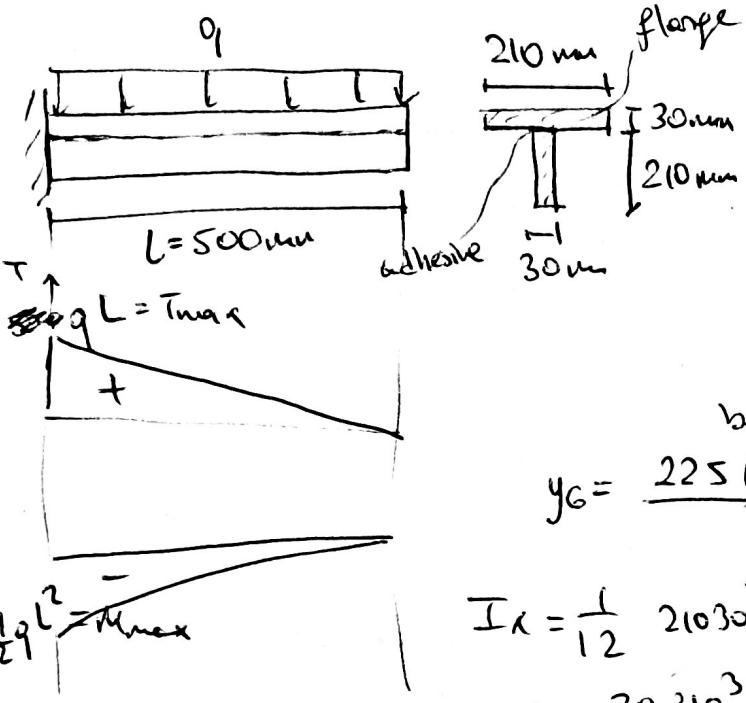
$q \rightarrow$ shear flow : shear force per unit length

$$q_{avg} = -\frac{T_{avg} \cdot S_x}{I_x} \quad \text{(average shear flow)}$$



$$T_{avg} = \frac{\int_z^{z+a} T_y(z)\, dz}{q}$$





The flange of wooden T section is connected to the web by adhesive.

$(\tau_{em})_{adh} = 2\ MPa$

$(\sigma'_{em})_{wood} = 8\ MPa$

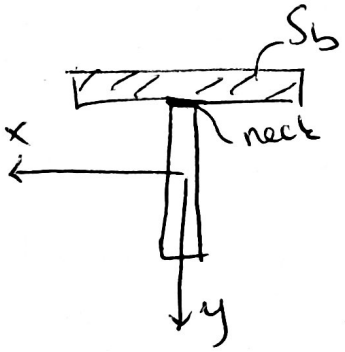
Find the maximum q value that beam can carry safely.

$$y_G = \frac{225(210\times30) + 105(210\times30)}{2(210\times30)} = 165\ mm$$

$$I_x = \frac{1}{12}21030^3 + 210.30(210+15-165)^2 + \frac{30\,210^3}{12} + 210.30(105-165)^2 = 6900\,10^4\ mm^4$$

check for shear at transition region

$$S_b = -210.30.(210+15-165) = -378\,10^3\ mm^3$$



$$\frac{-\tau_{max} S_b}{I_x t} \le (\tau_{em})_{adh} \quad \frac{-500q(-378\,10^3)}{6900\,10^4.30} \le 2$$

$$q \le 21.8\ N/mm$$

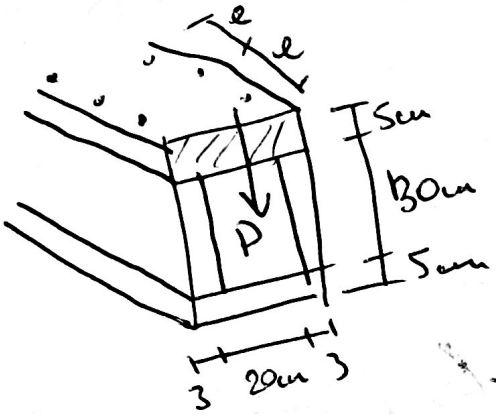
check for bending at the bottom of the section (max. stress - compressive)

$$\frac{M_{max}}{I_x} y_{max} \le (\sigma'_{em})_{wood} \quad \frac{(\frac{1}{2}q500^2)165}{6900\,10^4} \le 8$$

$$q = 26.75\ N/mm$$

$$q_{max} = 21.8\ N/mm$$

Ex:



Each sraws shown in figure can sustain 1 kN shear force. Taking into account bending with shear effect, determine the distance e between to successive screws if a load P=6kN is applied to the cross section.

$$I_x = \frac{26 \cdot 40^3}{12} - \frac{20 \cdot 30^3}{12} = 93666{,}67 \text{ cm}^4$$

$$S_b = -5 \cdot 26 \cdot 17{,}5 = -2275 \text{ cm}^3$$

$$q = -\frac{P \cdot S_b}{I_x} = \frac{-6 \cdot -2275}{93666{,}67} = 0{,}145 \text{ t}$$

$$\therefore q \cdot e = \underset{\text{number of screws}}{2} \cdot 1$$

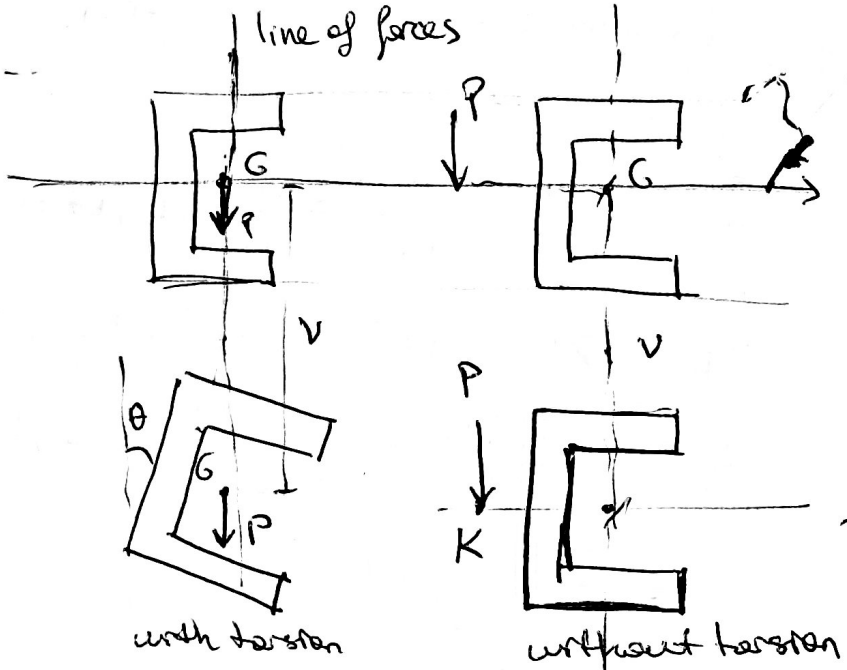
$$0{,}1457 \cdot e = 2 \cdot 1$$

$$e = 13{,}72 \text{ cm}$$

BENDING WITH AND WITHOUT TORSION

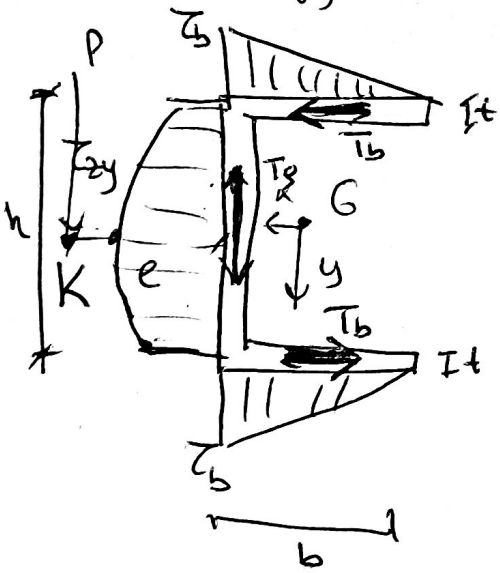
1 week

In the case of bending with shear, if the cross-section of a beam is not symmetric with respect to the line of forces which pass through the center of gravity, bending and torsion occurs together.



Let's find a point such that the shear force acts this point and does not cause torsion in the cross-section. This point is called the "Shear Center".

To clarify the concept, consider the section given below.



The forces which are the resultant of τ_{zx} occur on the horizontal arms are equal and opposite, It

$$T_b = \frac{1}{2}\tau_b\, b.t = \frac{1}{2}\left(-\frac{PS_b}{I_x t}\right).b.t \quad \text{where;} \quad S_b = \frac{1}{2}h\,b\,t$$

$$= \frac{h\,t\,b^2}{4 I_x t}$$

from vertical equilibrium

$$T_b = P$$

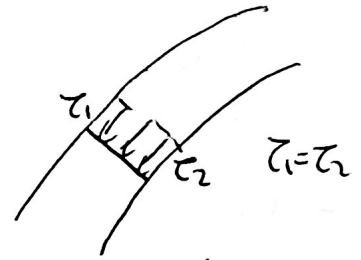
~~The total moment about 2 axes is~~

~~Either~~ $P.e = T_b.h = 0$

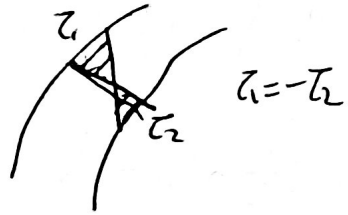
$$\boxed{e = \frac{h^2 b^2 t}{4 I_x}}$$

SHEAR CENTER FOR OPEN SECTIONS

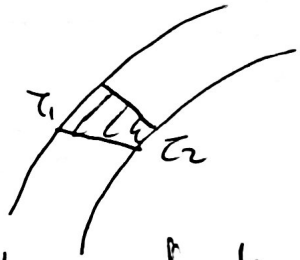
In the case of bending with torsion both shear stress and normal stress distributions are changed. Hence; for unsymmetrical section a way should be found sketch that no torsion occurs



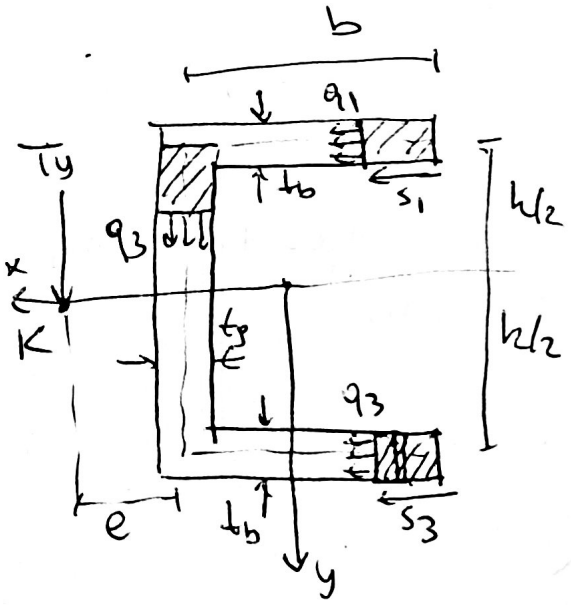
Bending without torsion



only torsion



torsion + bending



$$q_1 = \tau_{zx} t_b = -\frac{T_y (S_x)_1}{I_x}$$

$$q_2 = \tau_{zy} t_\varphi = -\frac{T_y (S_x)_2}{I_x}$$

$$q_3 = \tau_{zx} t_\varphi = -\frac{T_y (S_x)_3}{I_x}$$

$$(S_x)_1 = s_1 t_b \left(-\frac{h}{2}\right) = -\frac{1}{2} s_1 t_b h$$

$$(S_x)_2 = b t_b \left(-\frac{h}{2}\right) + s_2 t_\varphi \left(-\left(\frac{h}{2} - \frac{s_2}{2}\right)\right)$$

$$= -\frac{1}{2}\left(b t_b h + t_\varphi (h s_2 - s_2^2)\right)$$

$$(S_x)_3 = s_3 t_b \frac{h}{2} = \frac{1}{2} s_3 t_b h$$

Shear forces on the section arms

$$T_1 = \int_0^b q_1 ds_1 = -\frac{T_y}{I_x} \int_0^b \left(-\frac{1}{2} s_1 t_b h\right) ds_1 = \frac{T_y (b^2 t_b h)}{4 I_x}$$

$$T_3 = \int_0^b q_3 ds_3 = -\frac{T_y}{I_x} \int_0^b \frac{1}{2} s_3 t_b h \, ds_3 = -\frac{T_y (b^2 t_b h)}{4 I_x}$$

Shear force in the web

$$I_x = 2\left[\frac{1}{12} b t_b^3 + b t_b \left(\frac{h}{2}\right)^2\right] + \frac{1}{12} t_p h^3$$

$$= \frac{1}{6} b t_b^3 + \frac{1}{2} b t_b h^2 + \frac{1}{12} t_p h^3 \simeq \frac{1}{2} b t_b h^2 + \frac{1}{12} t_p h^3$$

$$T_2 = \int_0^h q_2 ds_2 = -\frac{T_y}{I_x} \int_0^h -\left[\frac{1}{2} b t_b h + \frac{1}{2} t_p \left(h s_2 - s_2^2\right)\right] ds_2$$

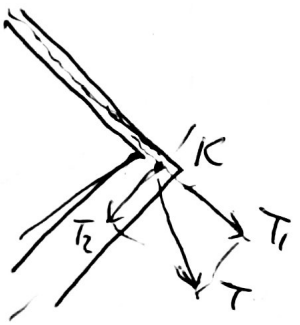
$$= \frac{T_y}{I_x}\left(\frac{1}{2} b t_b h^2 + \frac{1}{12} t_p h^3\right) = T_y$$

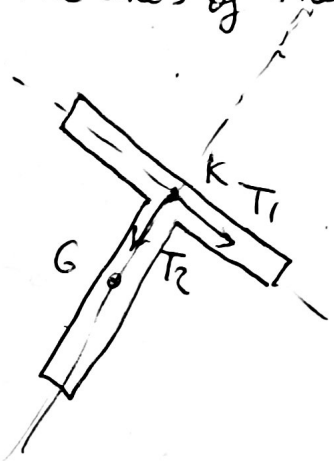
Determination of shear center:

$$+\circlearrowleft \sum M_K = 0 \qquad T_2 e - \frac{h}{2} T_1 + \frac{h}{2} T_3 = 0$$

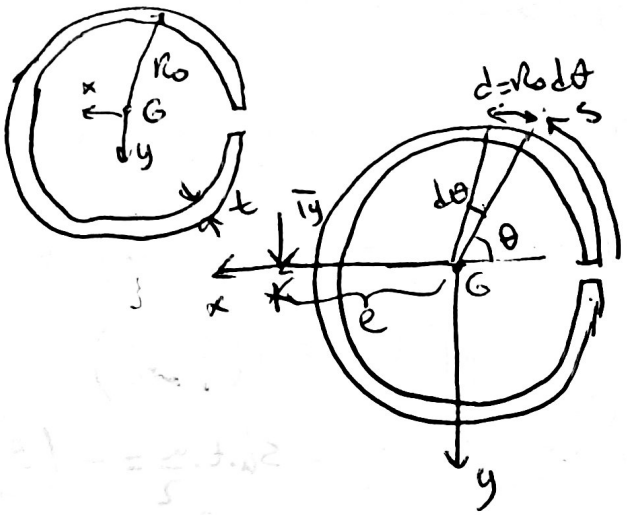
$$e = \frac{h^2 b^2 t_b}{4 I_x}$$

The shear center of the cross sections formed by two thin arms is the point of intersection of the axes of the arms





Ex Find the shear center?



$d = R_0 d\theta$

$dA = t ds = t R_0 d\theta$

$y = -R_0 \sin\theta$

$$S_x = \int_A y\, dA$$

$$= \int_0^{\theta} -R_0 \sin\theta \cdot t R_0 d\theta$$

$$= -t R_0^2 \int \sin\theta\, d\theta$$

$$= -t R_0^2 (1-\cos\theta)$$

$$I_x = \int_A y^2 dA$$

$$= \int_0^{2\pi} (-R\sin\theta)^2 t R_0 d\theta$$

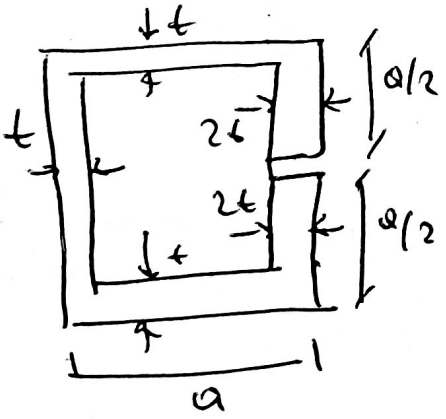
$$= t R_0^3 \int_0^{2\pi} \frac{1}{2}(1-\cos 2\theta) d\theta = \pi t R_0^3$$

Moment about G

$$T_y \cdot e = -\int_0^{2\pi} q\, ds\, R_0 = -\int_0^{2\pi} \frac{T_y S_x}{I_x} R_0\, ds$$

$$e = -\frac{1}{\pi t R_0^3} \int_0^{2\pi} R_0 \left(-t R_0^2 (1-\cos\theta)\right) R_0 d\theta = 2R_0$$

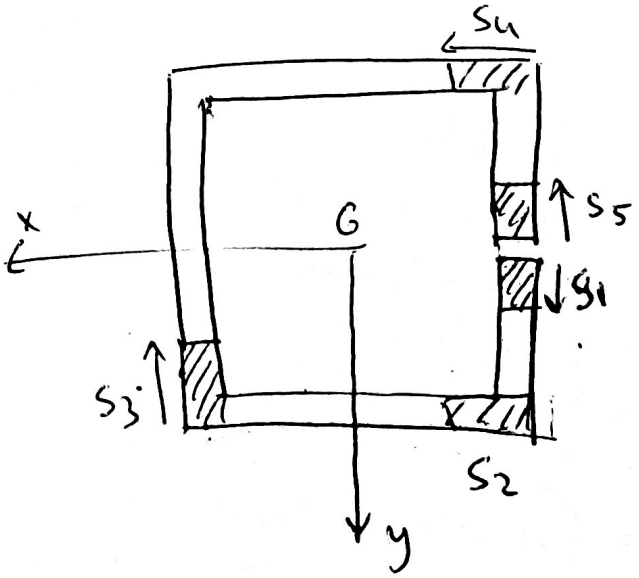
Ex



Find the shear center of open tube cross-section.

$$\bar{I}_x \cong \frac{1}{12} t a^3 + \frac{1}{12} 2t a^3 + at\left(\frac{a}{2}\right)^2 + at\left(\frac{a}{2}\right)^2$$

$$= \frac{3}{4} a^3 t \quad \text{(the terms includes } t^3 \text{ are ignored)}$$



$$S_x^1 = s_1 2t \cdot \frac{s_1}{2} = s_1^2 t$$

$$S_x^2 = +s_2 t \cdot \frac{a}{2} + \frac{a}{2} \cdot 2t \cdot \frac{a}{4} = \frac{s_2 a t}{2} + \frac{a^2 t}{4}$$

$$S_x^3 = t \cdot s_3 \left(\frac{a}{2} - \frac{s_3}{2}\right) + \frac{a}{2} \cdot 2t \cdot \frac{a}{4} + a t \cdot \frac{a}{2}$$

$$= \frac{3}{2} a^2 t + t s_3 \left(\frac{a}{2} - \frac{s_3}{2}\right)$$

$$S_x^4 = -\frac{a}{2} \cdot 2t \cdot \frac{a}{4} - s_4 \cdot t \cdot \frac{a}{2} = -\left(\frac{a^2 t}{4} + \frac{a t s_4}{2}\right)$$

$$S_x^5 = -s_5 \cdot 2t \cdot \frac{s_5}{2} = -s_5^2 t$$

shear flux on arms

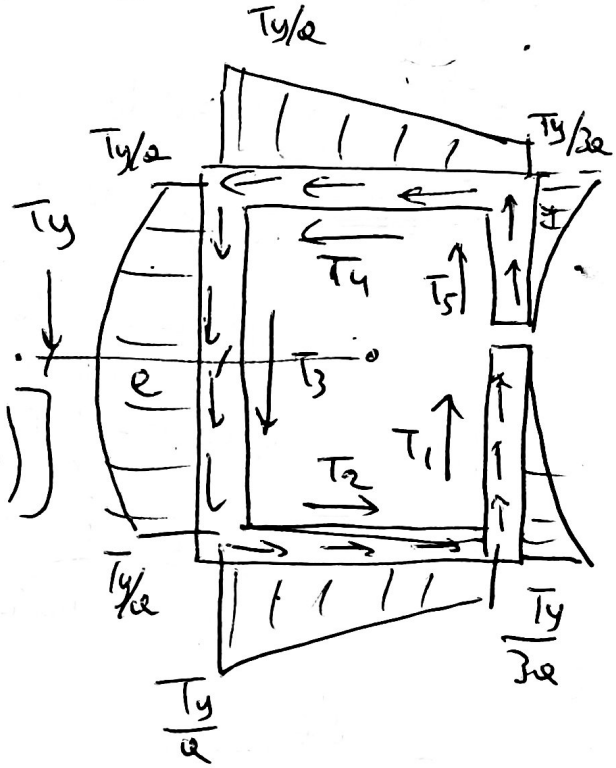
$$q_1 = \frac{-T_y (S_x)^1}{I_x} = \frac{T_y s_1^2 t}{3/4 a^3 t} = \frac{-4}{3} \frac{T_y s_1^2}{a^3}$$

$$q_2 = \frac{-T_y (S_x)^2}{I_x} = \frac{-T_y}{3a}\left(1 + \frac{2s_2}{a}\right)$$

$$q_3 = \frac{-T_y (S_x)^3}{I_x} = \frac{-T_y}{a}\left[1 + \frac{2}{3}\left(\frac{s_3}{a} - \frac{s_3^2}{a^2}\right)\right]$$

$$q_4 = -\frac{T_y (S_x)^4}{I_x} = \frac{T_y}{3a}\left(1 + \frac{2s_4}{a}\right)$$

$$q_5 = -\frac{T_y (S_x)^5}{I_x} = \frac{4 T_y s_5^2}{3a^3}$$



$$T_1 = \int_0^{a/2} q_4 \, ds_1 = \int_0^{a/2} -\frac{4}{3} \frac{T_y s_1^2}{a^3} ds_1 = \frac{-T_y}{18}$$

$$T_2 = \int_0^{a} q_2 \, ds_2 = \frac{-2}{3} T_y$$

$$T_3 = \int_0^{a} q_3 \, ds_3 = \frac{10}{9} T_y$$

$$T_4 = -T_2 = \frac{2}{3} T_y$$

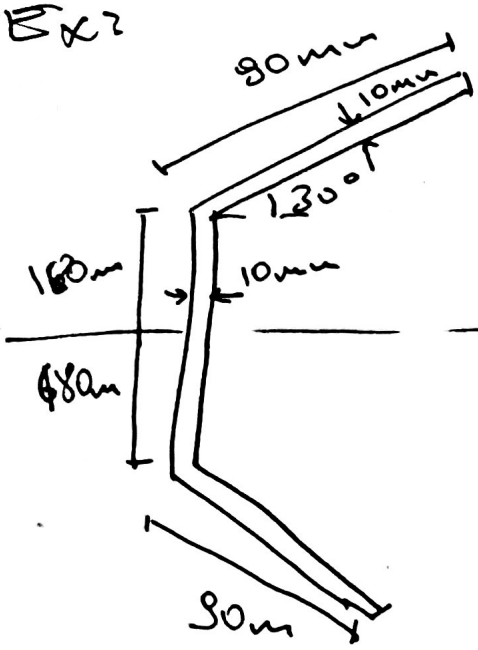
$$T_5 = -T_1 = \frac{T_y}{18}$$

Total moment about a point on the vertical arm is zero

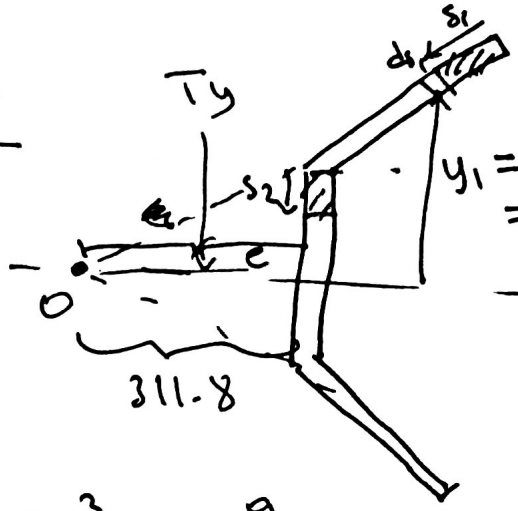
$$T_y \cdot e - T_5 \cdot a - T_1 \cdot a - T_4 \frac{a}{2} - T_2 \cdot \frac{a}{2} = 0$$

$$e = \frac{7}{9} a$$

Ex2



Find the location of shear center



$y_1 = 180 + (90 - s_1)\sin 30$

$= 225 - \frac{s_1}{2}$

$$I_x = \frac{t(2a)^3}{12} + 2\int_0^a y_1^2 (t\,ds)$$

$$= 3888\,10^4 + 2\left(506250 s_1 - 1125 s_1^2 + \frac{t}{5} s_1^3\right)\Big|_0^{90}$$

$$= 113\,10^6 \text{ mm}^4$$

$$S_x' = -s_1 t \cdot \left[180 + \left(90 - \frac{s_1}{2}\right)\sin 30\right]$$

$$= -2250 s_1 + 2.5 s_1^2$$

$$S_x^2 = S_x'\Big|_{s_1=90} + t s_2 \cdot \left[-\left(180 - \frac{s_2}{2}\right)\right]$$

$$= -182250 - 1800 s_2 + 5 s_2^2$$

$$q_1 = -\frac{T_y S_x'}{I_x} \qquad T_1 = \int_0^{90} q_1 ds_1 = -\frac{T_y}{I_x}\int_0^{90} -2250 s_1 + 2.5 s_1^2 \, ds_1$$

$$= -\frac{T_y}{I_x}\left(\frac{-2250 s_1^2}{2} + \frac{2.5 s_1^3}{3}\right)\Big|_0^{90} = 0.0753\, T_y$$

$$q_2 = -\frac{T_y S_x^2}{I_x} \qquad T_2 = \int_0^{360} q_2 ds_2 = -\frac{T_y}{I_x}\int_0^{360} \left(-182250 - 1800 s_2 + 5 s_2^2\right) ds_2$$

$$= -\frac{T_y}{I_x}\left(-182250 s_2 - \frac{1800 s_2^2}{2} + \frac{5 s_2^3}{3}\right)\Big|_0^{360}$$

$$= 0.8247\, T_y$$

Moment about O

$$T_y \cdot (311.8 - e) - 0.8247\, T_y \cdot 311.8 = 0$$

$$\boxed{e = 23.5 \text{ mm}}$$

Calculation shear flux when the coordinate axis is not principal

If the coordinat axis of the cross-section is not principal, then the normal stresses can be calculated by

$$\sigma_z = \frac{M_x\,(I_y \cdot y - I_{xy} x) + M_y\,(I_{xy} y - I_x x)}{I_x I_y - I_{xy}^2} \quad (1)$$

The shear flux is given as

$$\tau \cdot t = q = -\int_A \frac{d\sigma_z}{dz}\, dA \quad (2)$$

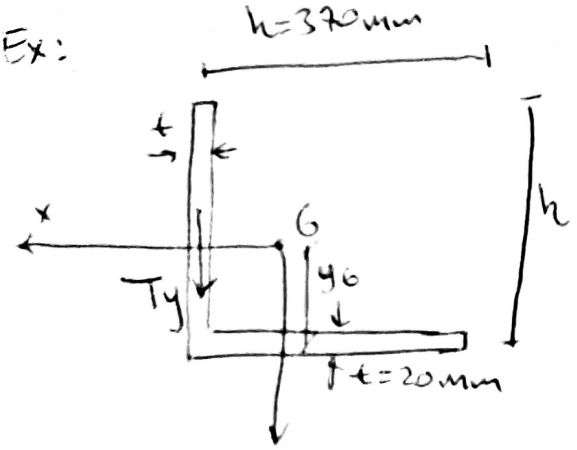
After substituting (1) into (2) and using relations $T_y = \frac{dM_x}{dz}$ one can obtain

$$\tau \cdot t = \frac{-1}{I_x I_y - I_{xy}^2} \int_{A'} \left[T_y\,(I_y \cdot y - I_{xy} x) - T_x\,(I_{xy} y - I_x x) \right] dA$$

Considering $S_x = \int y\, dA$ and $S_y = \int x\, dA$

$$q(s) = -\frac{1}{I_x I_y - I_{xy}^2} \left(T_y \left[I_y S_x(s) - I_{xy} S_y(s) \right] - T_x \left[I_{xy} S_x(s) - I_x S_y(s) \right] \right)$$

Ex:



Find the shear stress distribution on the given cross section

$$y_o = \frac{20 \cdot 370 \cdot (\frac{1}{2} 370)}{2(20\ 370)} = 92{,}5 \text{ mm}$$

$$x_o = 92.5 \text{ mm}$$

$$I_x = \frac{1}{12}(20 \cdot 370^3) + 20 \cdot 370 (-92.5)^2 + \frac{1}{12}(370 \cdot 20^3) + 370 \cdot 20 \cdot 92.5^2 =$$

$$= 2113\ 10^5 \text{ mm}^4$$

$I_y = 2113\ 10^5 \text{ mm}^4$ (due to

$$I_{xy} = (20 \cdot 370)\ 92{,}5\ (-92.5) + 370 \cdot 20\ (-92.5)\ 92.5 = -1266\ 10^5 \text{ mm}^4$$

direction of principal axis (x_1, y_1) $\tan 2\alpha_o = \frac{-2 I_{xy}}{I_x - I_y} = \infty$

$2\alpha_o = 90$ $\alpha_o = 45°$

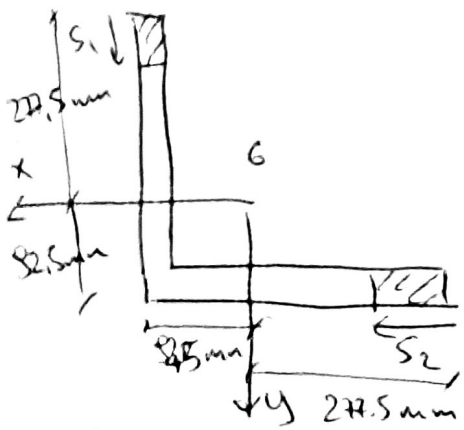
In general form

$$q = \tau \cdot t = -\frac{1}{I_x I_y - I_{xy}^2} \left(T_y \left[I_y S_x(s) - I_{xy} S_y(s) \right] - T_x \left[I_{xy} S_x(s) - I_x S_y(s) \right] \right)$$

since $T_x = 0$

$$q = \tau t = -\frac{T_y}{I_x I_y - I_{xy}^2} \left(I_y S_x - I_{xy} S_y \right)$$

$$I_x I_y - I_{xy}^2 = 2861\ 10^{13} \text{ mm}^8$$



$$S_{x1} = 20 s_1 \left[-(277.5 - \tfrac{1}{2} s_1) \right] = -5500 s_1 + 10 s_1^2$$

$$S_{y1} = 20 s_1\ 92.5 = 1850 s_1$$

$$S_{x2} = 20 s_2\ 92.5 = 1850 s_2$$

$$S_{y2} = 20 s_2 \left[-(277.5 - \frac{s_2}{2}) \right] = -5500 s_2 + 10 s_2^2$$

$$q_1 = -\frac{T_y}{I_x I_y - I_{xy}^2}\left(I_y S_{x_1} - I_{xy} S_{y_1}\right) = T_y\left(3280 s_1 - 7.38 s_1^2\right) 10^{-8}$$

$$q_2 = -\frac{T_y}{I_x I_y - I_{xy}^2}\left(I_y S_{x_2} - I_{xy} S_{y_2}\right) = T_y\left(1080 s_2 - 4.42 s_2^2\right) 10^{-8}$$

The max value of q_1 can be found as

$$\frac{dq_1}{ds_1} = 0 \quad \cancel{\Rightarrow 1080 - 4.42 s_2 = 0 \; (s_2}$$

$$s_1 = 222 \text{ mm}$$

The point in which q_2 [illegible]

$$1080 s_2 - 4.42 s_2^2 = 0 \qquad (s_2)_1 = 0 \qquad (s_2)_2 = 246 \text{ mm}$$

max value of q_2

$$\frac{\partial q_2}{\partial s_2} = 0 \Rightarrow s_2 = 12 \text{ mm}$$

Finally

$$q_1\Big|_{s_1=222} = 36.41\, 10^{-4} T_y \qquad q_2\Big|_{s_2=123} = 6.7\, 10^{-4} T_y$$

$$q_1\Big|_{s_1=370} = 20.26\, 10^{-4} T_y \qquad q_2\Big|_{s_2=370} = -20.26\, 10^{-4} T_y$$

